DESIGN DOCUMENT — LOCKED

SpotAwareStrategy

Sibling to DeterministicStrategy.
Driven by the +30.9pp aligned-vs-contrarian gap discovered in the 30-day Bonereaper analysis. Price-tier discipline borrowed from the leaderboard scout.

Polymarket 5-min BTC bot · Phase 5.7+ · Anti-vibe rule #1
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0. Why this strategy exists right now

Three pieces of evidence converged on 2026-05-08:

- The current **DeterministicStrategy fires zero times** on real 5-min BTC tick data at any threshold from 0.3pp to 0.6pp. The signal itself is too quiet.
- A different signal does have measurable edge: the 30-day Bonereaper spot-vs-book join (306,966 trades) shows **aligned trades win 67.4% vs contrarian 36.4%** — a +30.9pp gap that survives at scale.
- **High win rate without price-tier discipline loses money.** The two small-equity scout candidates had 96-98% scout WR and were both net-losing because they bought 95-99¢ shares that pay $1.

The spot-Δ signal + tier discipline + an EV check together describe a strategy whose expected value can be computed *before* it fires, not after.

1. The picture

```
      spot tick                book update
           \                      /
            v                    v
+-----------------------------------------+
|             SpotAwareStrategy           |
|                                         |
|  1. compute Δ_bps = 10000*(St-S0)/S0    |
|  2. classify alignment vs candidate side |
|  3. price-tier filter (reject ≥90% impl) |
|  4. lookup p_win from prior table       |
|  5. EV check: p_win*$1 > price + fees   |
|  6. fire BUY at $5 stake on aligned side |
|                                         |
+------------------+----------------------+
                   |
                   v
      Decision (state/decisions/.jsonl)
                   |
                   v
        PaperRunner sim-fill
                   |
                   v
        Dashboard side-by-side render
        vs DeterministicStrategy decisions
```

Sibling, not replacement. Both strategies receive the same MarketEvent stream; both write Decisions; the dashboard renders them in adjacent columns.

2. The signal: Δ_bps + alignment

For each 5-min Polymarket BTC market with window opening at `t = window_ts` and Binance BTCUSDT spot prices `S0` at window open and `St` at the time the bot considers entering:

```
Δ_bps = 10000 * (St - S0) / S0
```

Sign convention:

- Δ_bps > 0 ⇒ spot has moved UP since window open.
- Δ_bps < 0 ⇒ spot has moved DOWN since window open.
- Δ_bps ≈ 0 ⇒ spot is flat (|Δ|<5).

Alignment: UP side is aligned with Δ>0; DOWN side is aligned with Δ<0. The strategy **only** considers the aligned side. Contrarian and flat trades are off-table.

3. The trigger: thresholded + aligned

```
fire BUY on side X iff:
  X is aligned with sign(Δ_bps)          AND
  Δ_min ≤ |Δ_bps| ≤ Δ_max                AND
  price-tier filter passes (§4)           AND
  EV check passes (§5)
```

Δ_min default = 8 bps (midpoint of [5, 10] candidate space). **Δ_max ceiling = 50 bps** — past this point the data shows mean-reversion takes over and per-trade PnL flips negative. First paper-soak measures hit-rate and per-trade PnL across {5, 7, 8, 10} and locks the value at gate close.

4. The price-tier filter (lesson from the scout)

Reject any candidate where the entry price is in tier **85–100% implied probability**. Reason: Bonereaper's per-trade PnL in that tier is +$0.40 to +$0.65 before fees, which 40-60bps fee + slippage cuts to near zero.

Implied price	Tier	Use?	Why
< 0.15	<15%	DEFER	rare; +$8.83/trade but only 722 trades
0.15 – 0.30	15-30%	small sizing	aligned WR 25%, $+0.11/trade — thin
0.30 – 0.50	30-50%	YES — sweet	aligned WR 45.4%, $+1.17/trade
0.50 – 0.70	50-70%	YES — sweet	aligned WR 63.1%, $+0.95/trade
0.70 – 0.85	70-85%	cautious	aligned WR 78%, $+0.40/trade
0.85 – 1.00	85-100%	NO	aligned WR 91.7%, $+0.65/trade — fees eat it

The two sweet-spot tiers (30-50% and 50-70%) account for **$55.2k of the $83.1k aligned-trade gross PnL** in Bonereaper's 30-day data — 66% of the alignment edge in 53% of the alignment trades. That's where our trigger lives.

5. The EV check

```
p_win  = lookup table value (§6) for (Δ_bps bucket, price tier)
fees   = 0.005 (Polymarket gasless relayer estimate)
edge   = p_win * 1.0  -  (entry_price + fees)
fire iff edge > 0.005   # > 0.5¢ EV per $1 of notional
```

This single check is the difference between Bonereaper (winner) and Slow-Minute (-$3,964 / 7d). Slow-Minute fired on alignment but never gated on price; bought $0.99 shares and lost the spread. Our strategy gates explicitly.

6. Estimated p_win lookup table

Initial values from the 30-day Bonereaper bucket data; updated weekly from our own paper-soak data, never overwritten without a ≥200 trades-per-bucket sample. If a (tier, Δ) cell is missing, the strategy refuses to fire there. Conservative-by-default.

Price tier	Δ_bps bucket	Initial p_win	Source-trades
30-50%	5-10 bps	0.65	1,200
30-50%	10-20 bps	0.72	950
30-50%	20-50 bps	0.78	380
50-70%	5-10 bps	0.74	6,300
50-70%	10-20 bps	0.81	2,100

50-70%	20-50 bps	0.85	720
70-85%	5-10 bps	0.80	5,200
70-85%	10-20 bps	0.86	1,900
70-85%	20-50 bps	0.88	600
15-30%	5-10 bps	0.40	200
15-30%	10-20 bps	0.45	130

7. Architecture: sibling strategy in PaperRunner

```
src/polymarket_bot/strategies/
  __init__.py
  protocol.py            (existing)
  deterministic.py       (existing)
  ensemble.py            (existing — for the LLM ensemble)
  spot_aware.py          (NEW — this strategy)
  spot_aware_priors.py   (NEW — loader + bucket lookup)
data/
  spot_aware_priors.json (NEW — versioned prior table)
```

Both strategies share the Strategy protocol. PaperRunner is extended to host strategies: list[Strategy]; both receive every event; each Decision is tagged with its strategy name. Dashboard groups by that field and renders side-by-side.

8. Sizing

$5 flat per trade for the first 100 trades (locked decision #9). No Kelly logic in this strategy. After 100 paper trades we have a sample to gate against; THAT is when fractional-Kelly enters the picture. Sizing logic lives in the executor layer (Phase 6), not this strategy.

9. Paper-soak gate criteria — REALISTIC

This is where the operator's $50–$250/day target gets reconciled with $1k-bankroll one-market math. The criteria below are what the strategy must clear in a 7-day paper-soak before any move toward live trading.

Set A — the strategy works:

#	Metric	Target	Why
A1	Fires per day	30 ≤ fires ≤ 200	Sanity-check selectivity
A2	Net win rate	≥ 55%	Beats coin-flip with margin
A3	Aligned-vs-contrarian gap	≥ +20pp	≥ ■ of Bonereaper's 30d gap
A4	Per-trade NET PnL	≥ +$0.20	After 0.5% fee + slippage
A5	Net daily PnL	≥ +$5.00	A1 × A4 lower bound = $6/day
A6	Max drawdown	≤ 15% of bankroll	Hard ceiling; halt + review

Set B — the infrastructure works:

#	Metric	Target
B1	Zero unhandled exceptions in 7d	match Phase 5.7 #7
B2	Litestream replicating without errors	match Phase 5.7 #8

B3	Decision write rate	≥ 1 per 12h log file
B4	EV-rejection-vs-fire ratio	log it, don't gate

Why $5/day is the gate, not $50: 288 5-min BTC windows × 30% fire rate = ~86 fires/day max. At $0.30 net per trade scaled to $5 stake (12.7% of Bonereaper's avg trade size) and a 2× selectivity boost from our tier filter, the math says $6.55/day NET on a perfect run. Anything materially higher means we got lucky on a sample, not that the edge is structural.

10. Backtest plan

Before the strategy ever runs in paper-soak, it must clear a backtest against captured ticks + 30 days of Binance BTCUSDT 1m klines (already pulled). For each (Δ_bps, price tier) cell the backtest's p_win should land within ±5pp of Bonereaper's bucket WR. Drift beyond that means signal decay or join error.

Backtest gate: if A1-A4 cannot be met on 30d of historical data, do NOT promote to paper-soak. Tighten Δ_min, expand the tier filter, or re-derive priors from a longer window. Backtest is a necessary but not sufficient gate; live microstructure (latency, partial fills, ladder shifts) still requires paper-soak.

11. Scaling roadmap to $50–$250/day NET

Aspirational, multi-phase. Phases 11.B onward are explicitly OUT OF SCOPE for this design doc and require their own design + gate. Listed here so the long-range target the operator wants is visible and tracked.

Phase	What changes	Bankroll	Markets	Realistic NET ceiling
11.A	$5 stake, 1 market BTC 5m, paper soak	$1k	1	$5–25/day
11.B	Real-money go-live; same stake/market	$1k	1	$5–25/day
11.C	Stake grows with bankroll; $10–25/trade	$2k–5k	1	$20–80/day
11.D	Multi-market: BTC 15m + 1h on top of 5m	$5k	3	$50–150/day
11.E	Cross-asset: ETH 5m/15m/1h	$7.5k	6	$100–250/day
11.F	Sports + calendar effects	$10k+	20+	$200–500/day

Each phase has its own gate. **$50/day is plausible at Phase 11.D — only after** 11.A clears the +$5/day paper gate, 11.B clears the same gate live with real fills, and 11.C demonstrates that bankroll-scaled sizing doesn't break the per-trade EV math.

12. What this doc does NOT do

- Does **not** un-archive locked decision #19 (Bonereaper cloning). We are extracting one signal + one discipline, not cloning.
- Does **not** lower locked decision #6's threshold. DeterministicStrategy keeps its 0.6pp threshold.
- Does **not** turn on the LLM ensemble. Decisions #5 and #16 stand.
- Does **not** commit to live trading. Phase 6 still gated behind Phase 1B + this strategy's paper-soak gate.
- Does **not** write code in this commit. Code lands in a follow-up commit that imports this doc as the spec.

13. Open questions / icebox

- Bucket boundary for <15% tier — defer until ≥1k paper-soak trades validate the +$8.83/trade headline.
- Δ_min by time-of-day — log it, study after 7d soak.
- Spot source: Binance for now; don't change while priors are derived from it.

- Fee model: 0.5% placeholder; measure on real fills at Phase 6 prereq.
- Halt sentinel respect: strategy must check state/halt.sentinel before firing.
- Decision schema field: confirm 'strategy: str' is on Decision before code lands.

14. Implementation plan (next session)

In order, with a failing test before each piece per anti-vibe rule #2:

Step	What	LOC est.
1	tests/strategies/test_spot_aware.py — failing tests	~120
2	data/spot_aware_priors.json — initial table	~50
3	spot_aware.py + spot_aware_priors.py — implementation	~250
4	_paper/runner.py — host two strategies	~50
5	scripts/run_backtest.py --strategy=spot_aware	~30
6	runbooks/gates/spot-aware-paper.sh — 7d audit script	~80
7	Deploy to VPS, leave running 7 days, dashboard glue	—
8	Run gate audit, tag spot-aware-paper-pass on green	—

Estimated LLM cost: **$0**. Wall-clock: ~3h code + 7d soak.

Appendix — quick-reference math

Why $5/day net is the gate, not $50:

```
Bonereaper 30d data:
  306,966 trades × $12.13M notional × 1.94% gross ROI = $234,864 gross
  Net WR = 51.7%; per-trade gross = $0.76; per-trade net ≈ $0.30

Our scale:
  $5 stake / $39.41 Bonereaper avg = 12.7% of their per-trade size
  288 windows/day × 30% fire rate = 86 fires/day max
  86 × $0.30 × 12.7% = $3.28/day on Bonereaper's blended ROI

Selectivity boost (we filter heavy favorites):
  86 × $0.30 × 12.7% × 2 = $6.55/day  ⇒  rounded gate = $5/day NET
```

Why $50–$250/day appears at scale, not at start:

To 10× from $5/day NET to $50/day NET, we need any combination of: (a) 2× per-trade EV from better priors after months of paper data, (b) 2× trade volume from BTC 15m + 1h, (c) 2.5× stake size from a Kelly-sized larger bankroll. Each is a separate design + gate. None are in this doc's scope. The path is visible — that's the operator's actual ask.